



Supermarket store branches bold text sales analysis

Introduction

This project performs Exploratory Data Analysis (EDA) on a Store Sales dataset using Python and Google Colab. The dataset contains information about store area, items available, daily customer count, and store sales. The main objective of this project is to analyze business performance, customer behavior, and sales trends using statistical analysis and visualization techniques.

Problems Identified in the Dataset

- Difficulty in understanding sales trends from raw data
- Lack of proper visualization for business analysis
- Challenges in identifying relationships between store area, customers, and sales
- Large numerical data without meaningful insights
- Need for better business reporting and analysis

Problems Solved During the Project

- Cleaned and processed the dataset using Python and Pandas
- Analyzed store sales and customer behavior
- Identified relationships between business variables
- Created charts and graphs for better understanding
- Used visualization techniques to identify patterns and trends

What I Learned From This Project

- Data Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- Business Data Analysis
- Data Visualization using Matplotlib and Seaborn
- Correlation and Trend Analysis
- Real-world dataset handling using Google Colab and Python

Recommended Data Policies

- Maintain accurate and organized business records
- Regularly update customer and sales information
- Remove duplicate or incorrect data entries
- Use structured datasets for better analysis
- Apply visualization techniques for business decision-making

Overall, this project improved practical knowledge of business analytics, visualization, and real-world sales data analysis using Python.

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

Dataset Overview

The Store Sales dataset contains information about store area, items available, daily customer count, and store sales. This dataset was analyzed using Exploratory Data Analysis (EDA) techniques to understand sales trends, customer behavior, and relationships between business variables through statistical analysis and data visualization.

```
In [2]: df = pd.read_csv("/content/Stores.csv")
```

What are the first five rows of the dataset?

```
In [3]: df.head()
```

```
Out[3]:
```

	Store ID	Store_Area	Items_Available	Daily_Customer_Count	Store_Sales
0	1	1659	1961	530	66490
1	2	1461	1752	210	39820
2	3	1340	1609	720	54010
3	4	1451	1748	620	53730
4	5	1770	2111	450	46620

What are the last five rows of the dataset?

```
In [5]: df.tail()
```

```
Out[5]:
```

	Store ID	Store_Area	Items_Available	Daily_Customer_Count	Store_Sales
891	892	1582	1910	1080	66390
892	893	1387	1663	850	82080
893	894	1200	1436	1060	76440
894	895	1299	1560	770	96610
895	896	1174	1429	1110	54340

What are the column names in the dataset?

```
In [6]: df.columns
```

```
Out[6]: Index(['Store ID ', 'Store_Area', 'Items_Available', 'Daily_Customer_Count',  
              'Store_Sales'],  
              dtype='object')
```

What is the shape of the dataset?

```
In [7]: df.shape
```

```
Out[7]: (896, 5)
```

What is the basic information of the dataset?

```
In [8]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 896 entries, 0 to 895  
Data columns (total 5 columns):  
#   Column                                Non-Null Count  Dtype  
---  ---                                -  
0   Store ID                             896 non-null    int64  
1   Store_Area                           896 non-null    int64  
2   Items_Available                       896 non-null    int64  
3   Daily_Customer_Count                 896 non-null    int64  
4   Store_Sales                          896 non-null    int64  
dtypes: int64(5)  
memory usage: 35.1 KB
```

What is the statistical summary of the dataset?

```
In [9]: df.describe()
```

```
Out[9]:
```

	Store ID	Store_Area	Items_Available	Daily_Customer_Count	Store
count	896.000000	896.000000	896.000000	896.000000	896.0
mean	448.500000	1485.409598	1782.035714	786.350446	59351.3
std	258.797218	250.237011	299.872053	265.389281	17190.7
min	1.000000	775.000000	932.000000	10.000000	14920.0
25%	224.750000	1316.750000	1575.500000	600.000000	46530.0
50%	448.500000	1477.000000	1773.500000	780.000000	58605.0
75%	672.250000	1653.500000	1982.750000	970.000000	71872.5
max	896.000000	2229.000000	2667.000000	1560.000000	116320.0

Are there any missing values in the dataset?

```
In [10]: df.isnull().sum()
```

```
Out[10]:
```

	0
Store ID	0
Store_Area	0
Items_Available	0
Daily_Customer_Count	0
Store_Sales	0

dtype: int64

What is the average store sales value?

```
In [14]: df["Store_Sales"].mean()
```

```
Out[14]: np.float64(59351.30580357143)
```

Which store has the highest sales?

```
In [15]: df.loc[df["Store_Sales"].idxmax()]
```

Out[15]:

	649
Store ID	650
Store_Area	1989
Items_Available	2414
Daily_Customer_Count	860
Store_Sales	116320

dtype: int64

Which store has the lowest sales?

In [16]: `df.loc[df["Store_Sales"].idxmin()]`

Out[16]:

	31
Store ID	32
Store_Area	1250
Items_Available	1508
Daily_Customer_Count	990
Store_Sales	14920

dtype: int64

What is the relationship between store area and store sales?

In [17]: `df[["Store_Area", "Store_Sales"]].corr()`

Out[17]:

	Store_Area	Store_Sales
Store_Area	1.000000	0.097474
Store_Sales	0.097474	1.000000

What is the relationship between daily customer count and store sales?

In [18]: `df[["Daily_Customer_Count", "Store_Sales"]].corr()`

Out[18]:

	Daily_Customer_Count	Store_Sales
Daily_Customer_Count	1.000000	0.008629
Store_Sales	0.008629	1.000000

What is the relationship between items available and store sales?

In [20]: `df[["Items_Available", "Store_Sales"]].corr()`

Out[20]:

	Items_Available	Store_Sales
Items_Available	1.000000	0.098849
Store_Sales	0.098849	1.000000

Store Area vs Store Sales Scatter Plot

Is there any relationship between store area and sales?

```
In [23]: import matplotlib.pyplot as plt

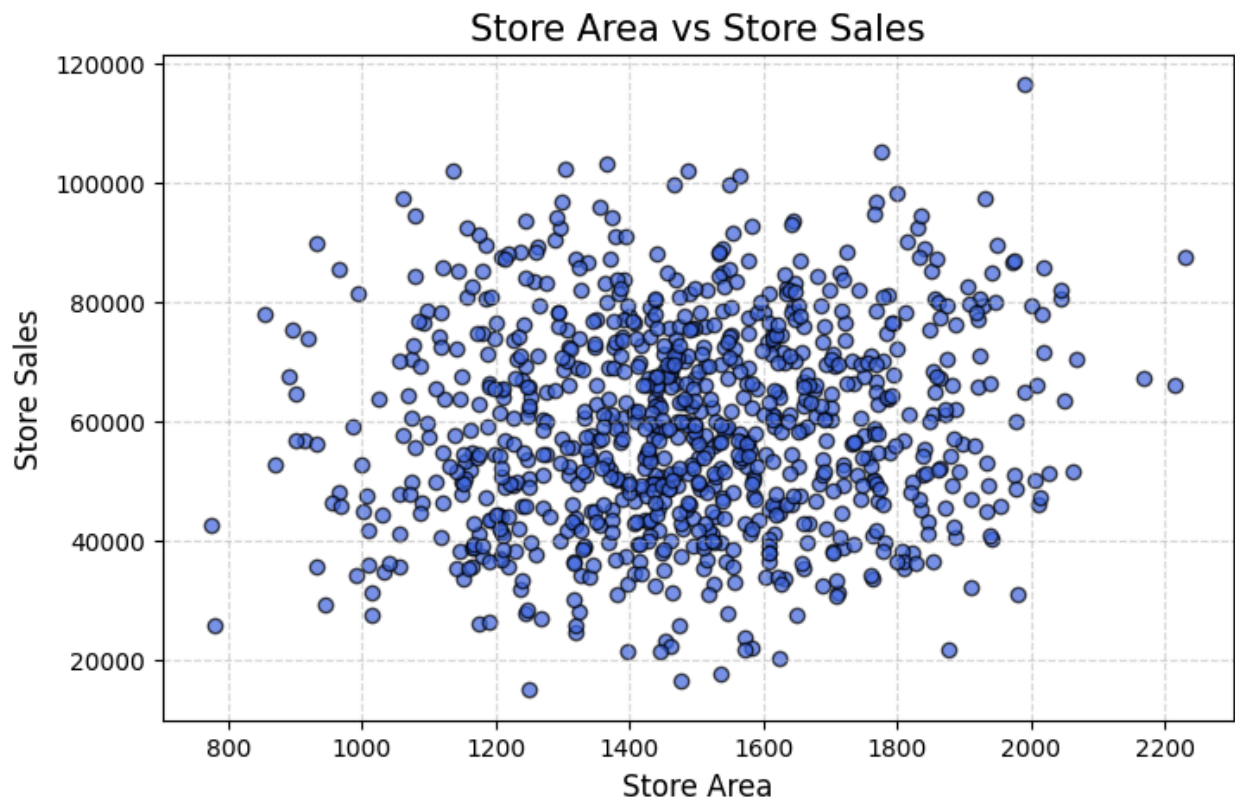
plt.figure(figsize=(8,5))

plt.scatter(
    df["Store_Area"],
    df["Store_Sales"],
    color="royalblue",
    edgecolor="black",
    alpha=0.7
)

plt.xlabel("Store Area", fontsize=12)
plt.ylabel("Store Sales", fontsize=12)
plt.title("Store Area vs Store Sales", fontsize=15)

plt.grid(True, linestyle="--", alpha=0.5)

plt.show()
```



Daily Customer Count vs Store Sales Scatter Plot

Is there any relationship between daily customer count and sales?

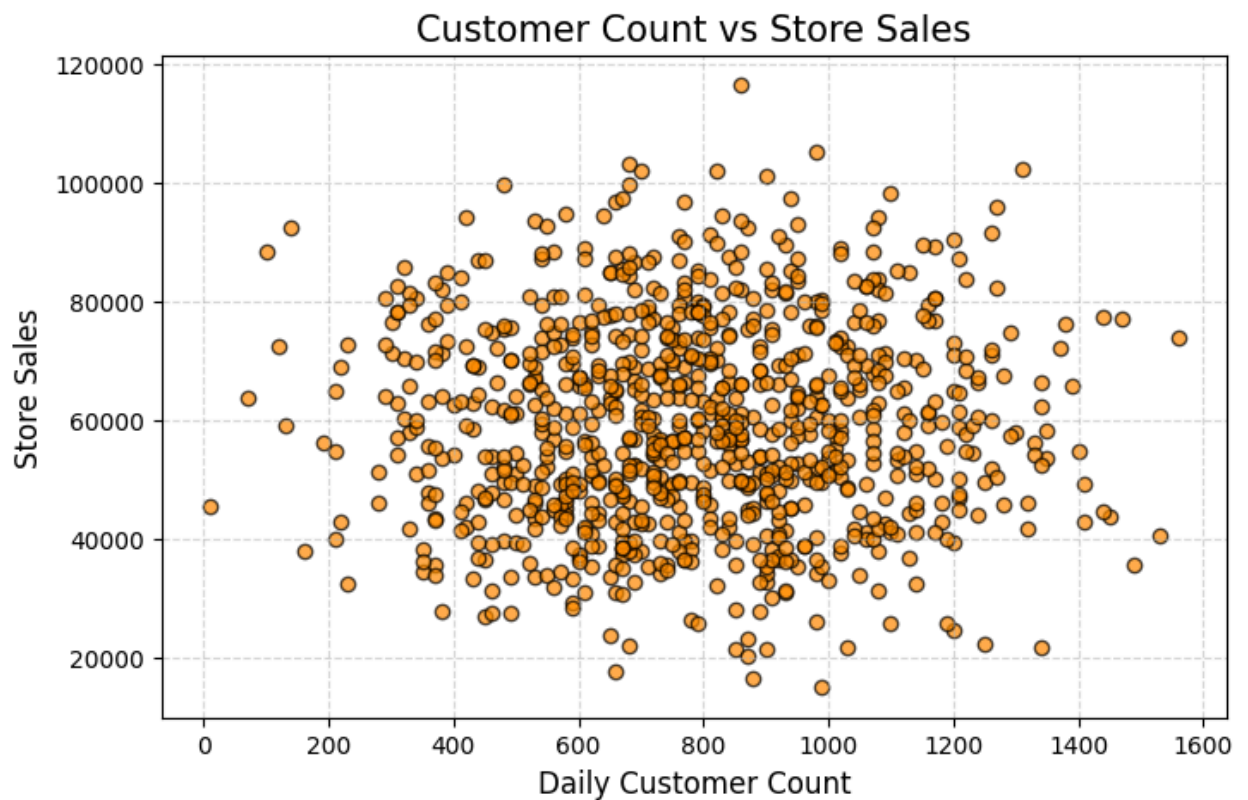
```
In [24]: plt.figure(figsize=(8,5))

plt.scatter(
    df["Daily_Customer_Count"],
    df["Store_Sales"],
    color="darkorange",
    edgecolor="black",
    alpha=0.7
)

plt.xlabel("Daily Customer Count", fontsize=12)
plt.ylabel("Store Sales", fontsize=12)
plt.title("Customer Count vs Store Sales", fontsize=15)

plt.grid(True, linestyle="--", alpha=0.5)

plt.show()
```



Items Available Distribution Histogram

What is the distribution of items available?

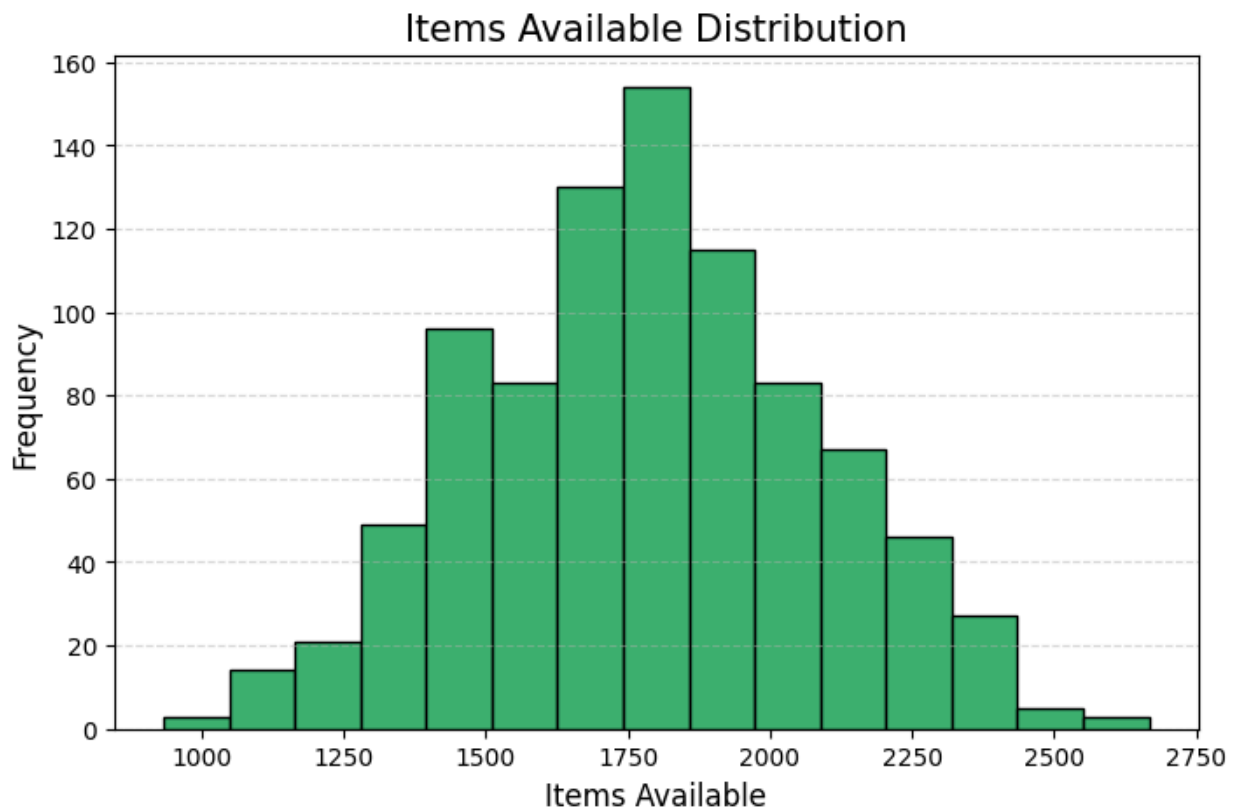
```
In [25]: plt.figure(figsize=(8,5))

plt.hist(
    df["Items_Available"],
    bins=15,
    color="mediumseagreen",
    edgecolor="black"
)

plt.xlabel("Items Available", fontsize=12)
plt.ylabel("Frequency", fontsize=12)
plt.title("Items Available Distribution", fontsize=15)

plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.show()
```



Store Sales Distribution Histogram

What is the distribution of store sales?

```
In [26]: plt.figure(figsize=(8,5))

plt.hist(
    df["Store_Sales"],
    bins=15,
    color="crimson",
    edgecolor="black"
)

plt.xlabel("Store Sales", fontsize=12)
plt.ylabel("Frequency", fontsize=12)
plt.title("Store Sales Distribution", fontsize=15)

plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.show()
```



Correlation Heatmap

What patterns can be observed from the correlation heatmap?

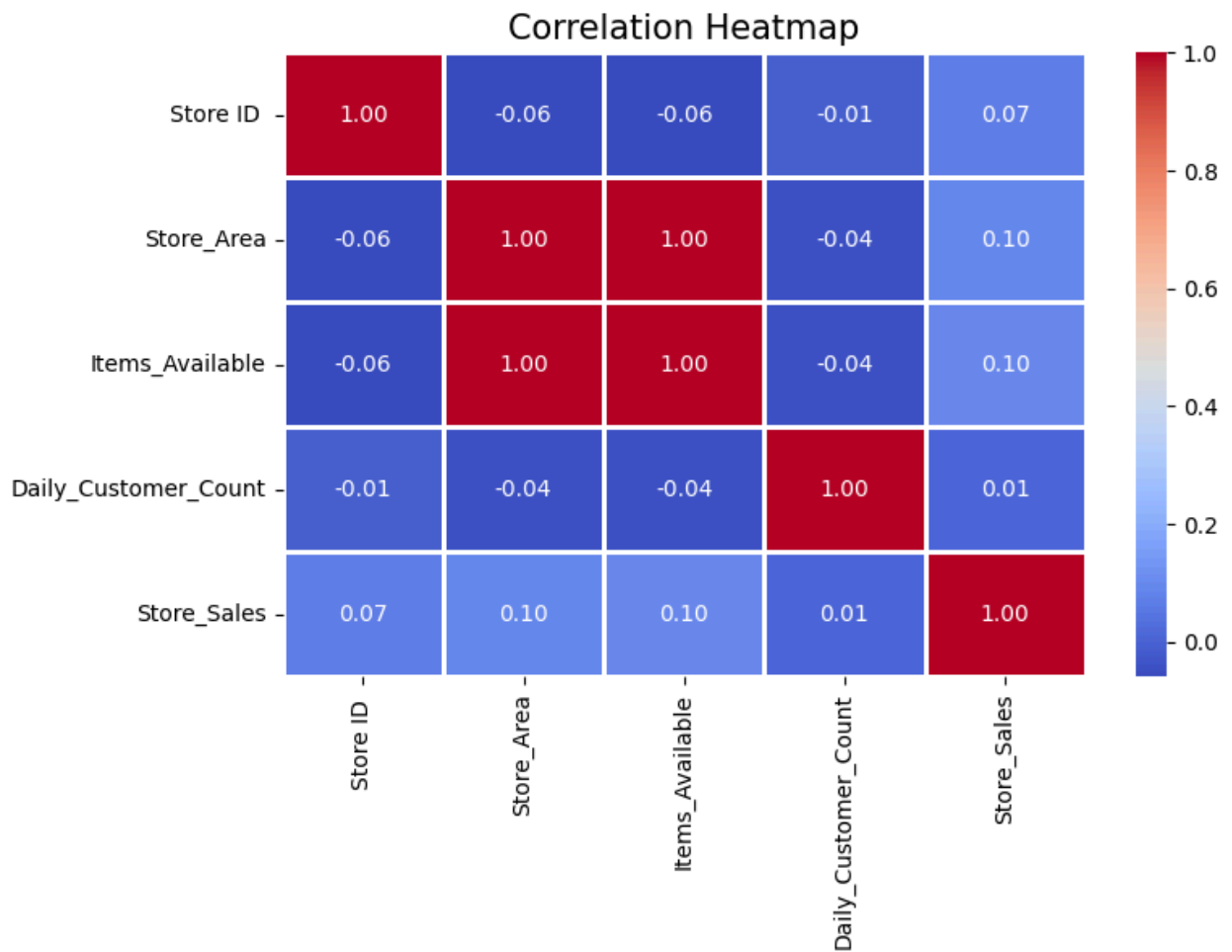
```
In [27]: import seaborn as sns

plt.figure(figsize=(8,5))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap="coolwarm",
    linewidths=1,
    fmt=".2f"
)

plt.title("Correlation Heatmap", fontsize=15)

plt.show()
```



Final Conclusion

This project successfully performed Exploratory Data Analysis (EDA) on the Store Sales dataset using Python and Google Colab. The analysis helped in understanding store performance, customer behavior, and sales trends through statistical analysis and visualization techniques.

Different business problems such as understanding sales patterns, customer activity, and relationships between variables were solved using data cleaning, preprocessing, and visualization methods. The project also improved practical knowledge of data analysis, business analytics, and real-world dataset handling.

Graphs Used in This Project

- Scatter Plot for Store Area vs Store Sales
- Scatter Plot for Customer Count vs Store Sales
- Histogram for Items Available Distribution

- Histogram for Store Sales Distribution
- Correlation Heatmap for Numerical Variables

Key Findings

- Store sales were influenced by store area and daily customer count
- Data visualization helped identify important business trends
- Correlation analysis showed relationships between numerical variables
- Histograms helped understand data distribution patterns
- Scatter plots helped analyze sales and customer relationships

Skills Improved During the Project

- Data Analysis
- Business Analytics
- Python Programming
- Data Visualization
- Exploratory Data Analysis (EDA)
- Real-world Dataset Handling

Overall Project Outcome

- Successfully analyzed store sales data using Python
- Generated meaningful insights from raw business data
- Improved understanding of customer behavior and sales trends
- Strengthened practical knowledge of EDA and visualization techniques